

Monitoring Amphibians in Small Water Bodies

Assessing methods and population status in the Czech Natura 2000 network

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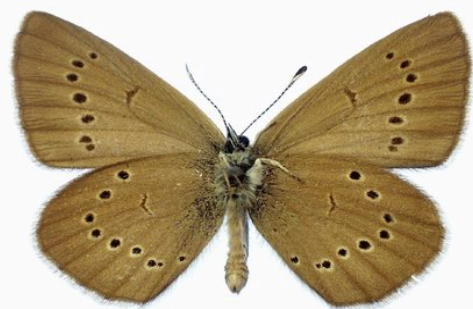
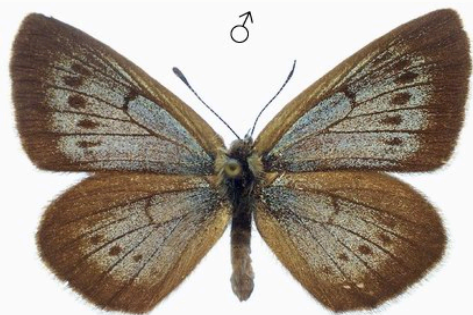
About me

Biodiversity Monitoring Specialist, Department of Biodiversity Monitoring, Nature Conservation Agency of the Czech Republic (NCA CR).

- **Background** – MSci Zoology (2022), supervised by Prof. Jason Matthiopoulos, University of Glasgow, UK
- **Current role** – supporting the digital transformation of conservation management through semi-automated site condition assessment within the **Nature Conservation Information System**
- **BiodivPond** – responsible for coordinating and leading the Biodiversa+ BiodivPond pilot
- **Experience I bring** – 6+ years of data analysis and management, sensor-based biodiversity monitoring, and international collaboration within **Biodiversa+ WP2**
- **Toolkit** – field surveys R spatial analysis long-term monitoring



MSci thesis – in brief



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Phengaris nausithous

Modelling the Effects of Resource Occurrence on the Metapopulation Dynamics of an Endangered Butterfly · 2022

- A butterfly whose life cycle depends on the foodplant *Sanguisorba officinalis* and *Myrmica* spp. host ants
- 126 habitat patches in the Křivoklátsko Protected Landscape Area, modelled as a metapopulation through patch persistence and colonisation
- A **Bayesian model fitted in JAGS**, built from the national **Species Occurrence Database** and **Habitat Mapping Database** rather than from field survey

Tools –

Bayesian hierarchical models

JAGS / MCMC

R · sf

spatial data



Proposed doctoral project

Monitoring methods for amphibian species of European importance in small water bodies and the assessment of their population status in the Czech Natura 2000 network



Aim and outputs

A **critical evaluation of traditional and novel methods** for monitoring the status of amphibian populations and habitats, including their **applicability for practical nature conservation**, and the **assessment of amphibian population status** in the Czech Natura 2000 network after seven years of standardised monitoring.

Aims

- Compare **detection success** of eDNA and PAM against conventional field methods
- Test their reliability for **screening habitat quality** and detecting Annex II flagship species
- Inform conservation on the **best methods for monitoring and management** of amphibian populations and habitats



Aim and outputs

A **critical evaluation of traditional and novel methods** for monitoring the status of amphibian populations and habitats, including their **applicability for practical nature conservation**, and the **assessment of amphibian population status** in the Czech Natura 2000 network after seven years of standardised monitoring.

Outputs

- **Scientific publications** on the method comparison
- **Recommendations** for biodiversity observation, and the **limits** of both approaches
- **Evaluation of amphibian status** in the Czech Natura 2000 network after seven years of standardised data



The methods compared

Conventional

- **Umbrella trapping** – amphibians, identification in the field
- **Sweep-netting** – amphibians and aquatic macroinvertebrates with centralised morphological identification by experts

Novel

- **eDNA metabarcoding** – amphibians, fish and aquatic macroinvertebrates from one sample
- **ddPCR** – targeted assay for *Triturus* spp.
- **PAM** – bats, and calling amphibians (*Pelophylax* spp., *Hyla* spp., *Bombina* spp.)

At the **60 BiodivPond core ponds**, amphibians and fish are surveyed by **eDNA including a quality check, *Triturus* spp. abundance estimation, and traditional sampling** – the same ponds, the same date.

An inter-laboratory ring test on identical eDNA samples runs alongside to establish consistency in extraction, amplification and species detection.



1 Method comparison

H1 Compared over a **single trapping event**, eDNA detects **most of the species present** better than umbrella traps.

H1b For **those species with similar detection rates**, ddPCR and umbrella trap results will **correlate** – and umbrella traps can therefore be used to **calibrate ddPCR**.

- Paired eDNA and traditional sampling at the same ponds, with **repeated traditional sampling** for *Triturus* spp. abundance
- The workplan already provides for **crested newt ddPCR** and a multimarker assay in the laboratory analysis
- Sites with both eDNA and expert data serve as testbeds for **detection accuracy, cost-effectiveness and data complementarity**



2 Seven years of standardised monitoring

H2 Both local and landscape factors play a significant role in amphibian population trends – with **management more important at smaller, more isolated sites.**

Local

- Management
- Hydrology
- Fish presence

Landscape

- Land cover
- Land use
- Connectivity

Testing this hypothesis will utilise data from 7+ years of a standardised monitoring scheme in the Czech Natura 2000 network. It will focus on **analysing the relationship between population trends, site characteristics, management practices, land cover and land use in the surrounding area.**



3 The BiodivPond pond network

H3 The role of **land cover** is **species-specific**.

H3b **Fish species associated with lower amphibian species richness** can be identified – eDNA returns the fish community from the same water sample.

- **560 ponds** sampled by eDNA – **60 core ponds** by active contributors and **500** through the citizen science scheme
- Across **different biogeographical regions of Europe**, temporary and permanent, with Natura 2000 sites prioritised but unprotected sites also included
- Biodiversity data are complemented by **pond nutrient status, vegetation coverage, surrounding land use and climatic variability**



BiodivPond – funding, data and network

Biodiversity monitoring of ponds – *evaluating the applicability of novel methods and identification of obstacles to wide use.* A Biodiversa+ species sub-pilot coordinated by **NCA CR**.

The data

- 560 ponds under one eDNA standardised protocol
- 60 ponds also covered by PAM and traditional sampling
- Centralised laboratory and bioinformatic analysis
- Open-access repository; data published via GBIF

The network

- Active contributors across at least 10 countries
- Dedicated lab, PAM and taxonomic analysis contributors
- Links to PONDERFUL, EuropaBON and other schemes

Total EC contribution **€1,083,899**, of which **€104,000** for coordination at NCA CR.



Prospective LIFE

A national-scale LIFE project coordinated by NCA CR, alongside which two species action plans relevant to this project are planned:

National species action plan

Epidalea calamita

- Active management of breeding sites
- Habitat restoration and creation
- Population monitoring and research: space for PAM, eDNA and traditional methods

Regional species action plan

Lissotriton montandoni – Beskydy

- Habitat restoration and creation
- Population monitoring and research: space for eDNA and traditional methods
- Research on terrestrial habitat use and connectivity

Both are instruments into which method recommendations and status assessments from this project would feed directly.



Thank you

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